------------------------- Slip 25 --------------------

Q.1) Write a C Program that demonstrates redirection of standard output to a file

[10 Marks]

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
   // Redirect stdout to "output.txt"
   FILE *fp = freopen("output.txt", "w", stdout);
   if (fp == NULL) {
      perror("freopen");
      return 1;
   }

   // All printf statements now go to output.txt
   printf("Hello, this will be written to output.txt\n");
   printf("File redirection is successful!\n");

   fclose(fp);
   return 0;
}
```

Q.2) Write a C program that redirects standard output to a file output.txt. (use of dup and open system call).

[20 Marks]

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
   int fd;
```

```c
    // Open (or create) the file output.txt for writing
    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0) {
        perror("open");
        return 1;
    }

    // Redirect stdout to fd
    if (dup2(fd, STDOUT_FILENO) < 0) {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd); // fd is no longer needed

    // Anything printed now goes to output.txt
    printf("This is written using dup2 and open system calls.\n");
    printf("Standard output has been redirected.\n");

    return 0;
}
```